ACP

Aarambh Chapterwise Problems

Class-10th

Control and Coordination

Control and Coordination

PART 1 (Basic MCQs)

Read the question properly!

1. Which hormone is secreted by the pancreas to regulate the blood sugar level?
(a) Thyroxin
(b) Insulin
(c) Adrenaline
(d) Growth hormone

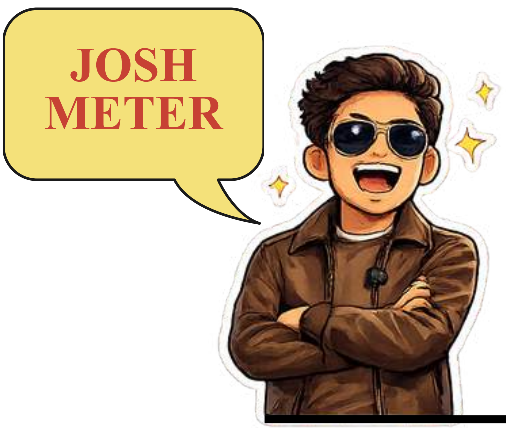
2. Spinal cord originates from
(a) cerebrum
(b) cerebellum
(c) medulla
(d) pons

3.. Which of the following correctly describes the function of receptors in our body?
(a) They carry instructions from the brain to muscles.
(b) They detect changes in the surroundings and generate nerve impulses.
(c) They control involuntary actions of the body.
(d) They release hormones directly into the bloodstream.

4. Select the correct pathway of reflex arc.
(a) Receptor → Effector → Afferent neuron → Efferent neuron → Association neuron
(b) Receptor → Efferent neuron → Association neuron → Afferent neuron → Effector
(c) Receptor → Association neuron → Afferent neuron → Effector
(d) Receptor → Afferent neuron → Association neuron → Efferent neuron → Effector

5. Which of the following is the main function of abscisic acid in plants ?
(a) To increase the length of cells
(b) To promote cell division
(c) To inhibit growth
(d) To promote growth of stem

6. Which of the following is a major function of auxin in plants?
(a) Closure of stomata
(b) Ripening of fruits
(c) Inhibition of plant growth
(d) Elongation of cells



7. Which hormone is not found in plants?
(a) Gibberellin
(b) Auxin
(c) Cytokinin
(d) Glucagon

8. Deficiency of growth hormone during childhood may result in:
(a) Diabetes
(b) Goitre
(c) Dwarfism
(d) High blood pressure

9. Salivary glands around the buccal cavity, gastric glands in the stomach wall and liver are some examples of
(a) heterocrine glands
(b) endocrine glands
(c) exocrine glands
(d) all of these.

10. What is the function of the pituitary gland?
(a) To develop sex organs in males
(b) To stimulate growth in all organs
(c) To regulate sugar and salt levels in the body
(d) To initiate metabolism in the body

11. Which of the following statements are correct?
(I) Hormones are released directly into the bloodstream.
(II) Endocrine glands use electrical impulses.
(III) Sex hormones regulate changes associated with puberty.
(a) (I) and (II)
(b) (I) and (III)
(c) (II) and (III)
(d) (I), (II) and (III)

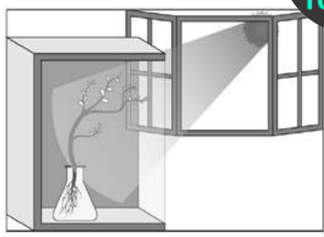
12. A female is suffering from an irregular menstrual cycle due to hormonal deficiency. Which hormone is most directly associated with regulation of the menstrual cycle?
(a) Oestrogen
(b) Testosterone
(c) Adrenalin
(d) Thyroxin

13. A pair of endocrine glands located in the human brain is:
(a) Parathyroid and Pituitary
(b) Pineal and Thymus
(c) Hypothalamus and Thymus
(d) Hypothalamus and Pineal.

14. The growth of pollen tubes towards ovules is due to:
(a) Hydrotropism
(b) Chemotropism
(c) Geotropism
(d) Phototropism

15. Raghav potted some germinated seeds in a pot. He put the pot in a cardboard box that was opened from one side. He keeps the box in a way that the open side of the box faces sunlight near his window. After 2-3

days, he observes the shoot bends towards the light, as shown in the image. Which type of tropism does he observe?

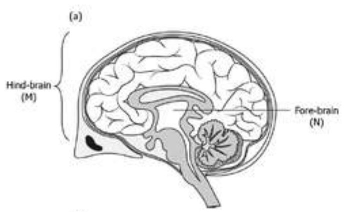


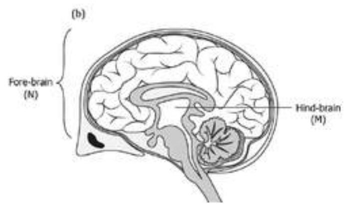
(a) Geotropism
(b) Phototropism
(c) Chemotropism
(d) Hydrotropism

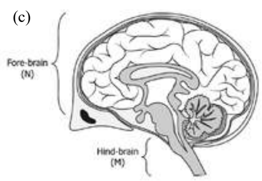
16. Which part of the brain controls blood pressure?
(a) Cerebrum
(b) Cerebellum
(c) Medulla oblongata
(d) Hypothalamus

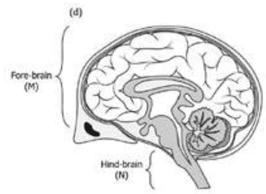
17. Organisms depend on hormones as well as electric impulses for the transmission of signals from the brain to the rest of the body. What can be a likely advantage of hormones over electric impulses?
(a) It is secreted by all types of cells present in the body.
(b) It is secreted by stimulated cells and reaches all cells of the body.
(c) It is relayed to the target organ faster than electric impulses.
(d) It does not depend on an external stimulus to be generated in the cells.

18. Which of the following option illustrates the location of the centre that controls the feelings associated with hunger (M) and the centre that allows a person to walk in a straight line (N)?

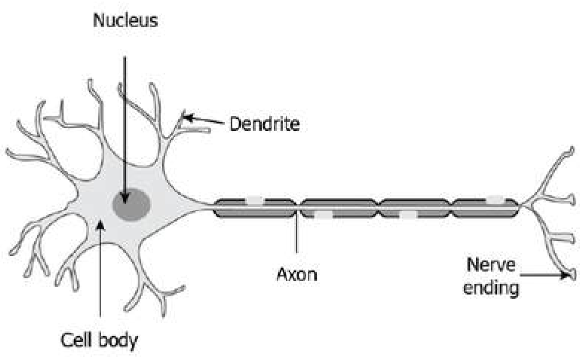








19. The image shows the structure of a neuron.



Which of the following options shows the mechanism of the travelling of sense in our body after our nose senses a smell?
(a) Olfactory receptor → Dendritic tip → Cell body → Axon → Nerve ending → Chemical signal released at the synapse → Dendrite of the next neuron
(b) Olfactory receptor → Axon → Cell body → Dendritic tip → Nerve ending → Next neuron
(c) Gustatory receptor → Dendritic tip → Cell body → Axon → Nerve ending → Dendrite of the next neuron
(d) Olfactory receptor → Cell body → Dendritic tip → Axon → Nerve ending → Next neuron

ASSERTION AND REASON TYPE QUESTION (1 Marks)

(yaha Marks katate h)

In the following questions a statement of Assertion is followed by a statement of Reason.
Mark the correct choice as two statements are given one labeled Assertion (A) and the other labeled Reason (R).
Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:
(a) Both A and R are true, and R is correct explanation of the assertion.
(b) Both A and R are true, but R is not the correct explanation of the assertion.
(c) A is true, but R is false.
(d) A is false, but R is true.

20. Assertion (A): A neuron transmits message in both directions.
Reason (R): Neuron is specialised for conducting information via electrical impulses from one part of body to another.

21. Assertion (A): Cerebellum controls the coordination of body movement and posture.
Reason (R): Medulla oblongata controls and regulates the centre for coughing, sneezing and vomiting.

22.Assertion (A): The movement of leaves of a sensitive plant is different from the movement of a shoot towards light.
Reason (R): The leaves of a sensitive plant show a non-directional nastic movement caused by changes in the amount of water in the cells, whereas the shoot shows a directional growth movement towards light.

23. Assertion (A): Adrenaline makes the heartbeat faster, resulting in supply of more oxygen to our muscles.
Reason (R): Adrenaline is secreted directly into the blood and carried to different parts of the body.

(ho jaayenge aaram se)

PART 2 (Subjective Questions)

SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)

24. Sunil unconsciously touched the hot iron rod and immediately withdrew his hand. Which type of action is involved? Also define this action. Draw a flow chart to show the path followed for this action.

25. "The timing and the amount of hormone secreted by a gland are regulated in the human body." Justify this statement with the help of an example.

26. Electrical impulse is an excellent means to transmit information in the animal body, but there are some limitations to the use of electrical impulses. State any two limitations.

27. A hormone 'X' is secreted in blood when a person is under scary situation.
(a) Identify the hormone 'X' and the gland that secretes it.
(b) Explain its role in dealing with scary or emergency situations.

28. What is the function of receptors in our body? What may happen if receptors fail to work properly?

29. (A) How is the movement of the leaves of a sensitive plant different from the downward movement of the roots? OR
(B) There is a hormone which regulates carbohydrate, protein and fat metabolism in our body. Name the hormone and the gland which secretes it. Why is it important for us to have iodised salt in our diet?

LONG ANSWER TYPE QUESTIONS (5 Marks)

(pahle points socho fir likho)

30. In life there are certain changes in the environment called 'stimuli' to which we respond appropriately. Touching a flame suddenly is a dangerous situation for us. One way is to think consciously about the possibility of burning and then moving the hand. But our body has been designed in such a way that we save ourself from such situations immediately.
(i) Name the action by which we protect ourself in the situation mentioned above and define it.
(ii) Write the role of (a) motor and (b) relay neuron.
(iii) (a) What are the two types of nervous system in the human body? Name the components of each of them.

OR

(iii) (b) Which part of the human brain is responsible for: (a) thinking (b) picking up a pencil (c) controlling blood pressure (d) controlling hunger

31. A person, while climbing up a rocky hill, comes into a panic state and fear. His body starts reacting in a "fight-or-flight" condition to adjust to the dangerous and stressful situation.
(a) (i) Name the hormone secreted in the blood of the person in this situation.

OR

(a) (ii) Name the source gland of the hormone secreted in this condition.
(b) State any two responses in the body of the person as a result of the secretion of this hormone.
(c) How does the action of the chemical signal in terms of hormones differ from the electrical impulses via nerve cells?

32. (a) (i) Distinguish between hormonal co-ordination in plants and animals.
(ii) Which part of the brain is responsible for - (1) Intelligence (2) riding a bicycle (3) vomiting (4) controlling hunger
(iii) How is the brain and spinal cord protected against mechanical injuries?

OR

(b) (i) What are tropic movements? Give an example of a plant hormone that (1) inhibits growth and (2) promotes cell division.
(ii) Explain the directional movement of a tendril in a pea plant in response to touch. Name the hormone responsible for this movement.

33. (a) (i) Define a reflex arc. Why have reflex arcs evolved in animals? Trace the sequence of events which occur, when you suddenly touch a hot object.
(ii) Name the part of the nervous system which helps in communication between the central nervous system and other parts of the body. What are the two components of this system?

OR

(b) (i) Leaves of 'chhui-mui' plant begin to fold up and droop in response to a stimulus. Name the stimulus and write the cause for such a rapid movement. Is there any growth involved in the movement?
(ii) Define geotropism in plants. What is meant by positive and negative geotropism? Give one example of each type.

34. (a) Name the hormone secreted by (i) Pituitary, and (ii) Thyroid stating one main function of each. Name the disorder a person is likely to suffer from due to the deficiency of the above mentioned hormones.
(b) How is the timing and amount of hormone released regulated ? Explain with an example.

35. How does chemical coordination occur in plants? Name four major plant hormones and state one function of each.

PART 3 (Advanced Questions)

Case Study/Source Based Question (5 Marks)

36. Read the source below and answer the questions that follow: During a sports event, Ananya had to catch a fast-moving ball. She was able to judge its speed, position, and direction and coordinate her hand movements to catch it. Her teacher explained that the human brain is responsible for these complex actions and different parts of the brain control different activities.
(a) Which part of the brain controls voluntary actions?
(b) Explain the functions of the cerebellum in maintaining body balance.
(c) What is the role of the spinal cord in nervous coordination?

OR

(c) What is the role of the cerebrum and cerebellum in catching the ball?

37. Read the source below and answer the questions that follow: Diya's uncle was diagnosed with diabetes and was advised to take insulin injections to control his blood sugar levels. Diya was curious about how hormones regulate body functions. Her teacher explained that hormones act as chemical messengers and control various activities in the body.
(a) What is the function of insulin in the human body?
(b) Explain how the secretion of hormones is regulated in the body.
(c) Why is iodine necessary in our diet?

OR

(c) Which hormone is responsible for the "fight or flight" response in humans?

38. Read the source below and answer the questions that follow: During a science experiment, Rohan placed his hand near a flame and quickly pulled it away. His teacher explained that this response occurred because of the transmission of nerve impulses through neurons. Neurons communicate with each other at special junctions called synapses, where electrical signals are converted into chemical signals.
(a) What is a synapse?
(b) Explain how nerve impulses are transmitted from one neuron to another.
(c) Why do nerve impulses travel in only one direction?

OR

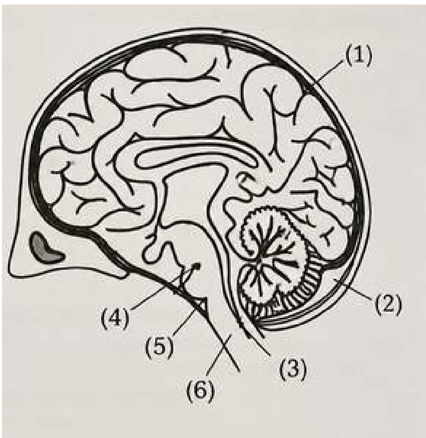
(c) Which part of a neuron receives the information and which part transmits it?

39. Read the source below and answer the questions that follow: Neha conducted an experiment where she placed a potted plant near a window. She noticed that over time, the plant's shoot bent towards the light. Her teacher explained that this response is due to the plant hormone auxin, which controls growth and movement.
(a) What is phototropism?
(b) How does auxin help in phototropism?
(c) What is geotropism? Give an example.

OR

(c) How do roots respond to water availability?

40. A child draws the diagram of human brain as shown:
(A) Identify the structure labelled as (2) and state its function.
(B) Identify the structure which has the following functions:
- Centre associated with hunger.
- Maintains posture and balance and ensures precision of voluntary movements.



Write one more function of such a structure other than given as clues.

(C) Write functions of structure marked as (6).

(D) Write differences between part (1) and (3) on the basis of:

- Their size
- Functions

(E) Identify the structure that makes the hind brain.

41. Mohan and Rohit observed that shoots of a plant growing in shade bend towards the sunlight. Whereas, leaves of 'Touch me not' plant fold and droop soon after touching. They were curious to know how these movements occur in plants.

In order to help them understand the movements in the plants, answer the following questions:

(A) What causes the bending of shoots in the plants as shown in figure A?

OR

What causes the folding of the leaves in 'Touch me not' plant as shown in figure B?

(B) Compare the movement of growth of the pollen tube towards ovule with the movement shown in part A of the above figure.

(C) Compare the movement shown in figure B with the movement of body parts in the animals.

- Neuron, Receptors & Transmission of Nerve Impulses – Q3, Q19, Q20, Q26, Q28, Q30, Q33, Q38
- Reflex Action & Reflex Arc – Q4, Q24, Q30, Q33
- Human Brain (Forebrain, Midbrain & Hindbrain) – Q2, Q16, Q18, Q21, Q30, Q32, Q36, Q40
- Endocrine Glands & Human Hormones – Q1, Q8, Q9, Q10, Q11, Q12, Q13, Q17, Q23, Q25, Q27, Q29(B), Q31, Q32(A), Q34, Q37
- Hormonal Coordination & Feedback Mechanism – Q17, Q25, Q31, Q32(A), Q34(B), Q35, Q37
- Plant Hormones (Auxin, Gibberellin, Cytokinin & ABA) – Q5, Q6, Q7, Q32(B), Q35, Q39, Q41
- Plant Movements (Tropic & Nastic Movements) – Q14, Q15, Q22, Q29(A), Q32(B), Q33(B), Q39, Q41
- Central & Peripheral Nervous System (CNS & PNS) – Q2, Q24, Q30, Q33(A), Q36
- Protection of Brain & Spinal Cord – Q32(A), Q40
- Adrenaline & Fight-or-Flight Response – Q23, Q27, Q31, Q37
- Neuron, Synapse & Synaptic Transmission – Q19, Q20, Q38
- Brain Functions (Cerebrum, Cerebellum, Medulla & Hypothalamus) – Q16, Q18, Q21, Q30, Q32, Q36, Q40
- Insulin, Thyroxin & Diabetes – Q1, Q25, Q29(B), Q34, Q37
- Types of Tropism (Phototropism, Geotropism, Hydrotropism, Chemotropism & Thigmotropism) – Q14, Q15, Q22, Q29(A), Q32(B), Q33(B), Q39, Q41
- Touch-me-not Plant (Nastic Movement) – Q22, Q29(A), Q33(B), Q41
- Communication and Coordination in Animals & Plants – Q3, Q17, Q19, Q20, Q26, Q28, Q31, Q32(A), Q35, Q38, Q39, Q41

SOLUTIONS

MULTIPLE CHOICE QUESTIONS [MCQ's]

1. (b)
2. (c)
3. (b)
4. (d)
5. (c)
6. (d)
7. (d)
8. (c)
9. (c)
10. (b)
11. (b)
12. (a)
13. (d)
14. (b)
15. (b)
16. (c)
17. (b)
18. (d)
19. (a)
20. (d)
21. (b)
22. (a)
23. (b)

SHORT ANSWER TYPE QUESTIONS:

24. A reflex action is a quick, automatic and involuntary response of the body to a stimulus. It takes place without conscious thinking and is mainly controlled by the spinal cord.

When Sunil touched the hot iron rod, heat acted as a stimulus. The receptors present in the skin detected the stimulus and immediately sent nerve impulses to the spinal cord through the sensory neuron. The spinal cord instantly sent impulses through the motor neuron to the muscles of the hand. As a result, the muscles contracted and the hand was withdrawn immediately without waiting for instructions from the brain. This helps prevent serious injury.

Flow Chart: Hot Iron Rod (Stimulus) → Receptors in Skin → Sensory Neuron → Relay Neuron (Spinal Cord) → Motor Neuron → Muscles of Hand (Effector) → Hand Withdrawn (Response)

25. Hormones are secreted only when required by the body. Their secretion is regulated by a feedback mechanism, which ensures that hormones are released in the correct amount and at the correct time.

Example: Regulation of Blood Sugar

- When the blood sugar level increases after eating food, the pancreas secretes the hormone insulin.
- Insulin helps body cells absorb glucose and lowers the blood sugar level.
- Once the blood sugar level becomes normal, the pancreas reduces or stops the secretion of insulin.
- Thus, both the timing and the amount of insulin secretion are controlled according to the body's needs.

This regulation helps maintain the normal functioning of the body.

26. Electrical impulses can reach only those cells that are connected by nervous tissue. Therefore, they cannot communicate with every cell of the body.

- Once a nerve cell generates and transmits an electrical impulse, it requires some time to reset before producing another impulse. Therefore, electrical impulses cannot provide continuous and steady communication.

27. (a) Hormone (X): Adrenaline

Gland: Adrenal gland

(b) Adrenaline helps the body prepare for emergency situations by:

- Increasing the heartbeat.
- Increasing the breathing rate.
- Increasing blood pressure.
- Increasing the blood glucose level by breaking down stored glycogen.

- Diverting more blood to the muscles.

These changes prepare the body for fight or flight, enabling a person to react quickly during emergencies.

28. Receptors are specialised cells present in sense organs that detect changes or stimuli in the surroundings and generate electrical impulses.
If receptors fail to work properly, the body may not detect the stimulus and may not respond appropriately. For example, damaged heat receptors may prevent a person from sensing a hot object, resulting in burns or injury.

29. (A)

Movement of Sensitive Plant Leaves	Downward Movement of Roots
It is a Seismonastic (Nastic) movement.	It is Positive Geotropism (Tropic movement).
It is caused by touch.	It is caused by gravity.
It is non-directional.	It is directional, towards gravity.
It occurs due to changes in turgor pressure in cells.	It occurs due to growth of cells.
It is temporary and reversible.	It is permanent and irreversible.

OR

29 (B). Hormone: Thyroxine
Gland: Thyroid gland
Importance of Iodised Salt:

- Iodine is essential for the thyroid gland to produce thyroxine.
- Thyroxine regulates the metabolism of carbohydrates, proteins and fats.
- It also helps in normal growth and development of the body.
- Deficiency of iodine leads to goitre, in which the thyroid gland enlarges, causing swelling in the neck.
- Therefore, iodised salt should be included in our diet to ensure proper production of thyroxine and prevent iodine deficiency disorders.

LONG ANSWER TYPE QUESTIONS:

30.(i) The action is called Reflex Action.
Definition: A reflex action is a quick, automatic and involuntary response of the body to a stimulus. It is controlled mainly by the spinal cord without involving conscious thinking.
(ii) Write the role of:
(a) Motor Neuron - Motor neurons carry nerve impulses from the spinal cord or brain to the effector organs (muscles or glands) to produce a response.
(b) Relay Neuron - Relay neurons are present in the spinal cord or brain. They connect sensory neurons with motor neurons and help transmit nerve impulses during reflex actions.
(iii)(a) What are the two types of nervous system in the human body? Name the components of each.
1. Central Nervous System (CNS)

- Brain
- Spinal Cord

2. Peripheral Nervous System (PNS)

- Cranial Nerves
- Spinal Nerves

OR

(iii)(b) (a) Thinking → Cerebrum
(b) Picking up a pencil → Cerebellum

(c) Controlling blood pressure → Medulla Oblongata
(d) Controlling hunger → Hypothalamus (Forebrain)

31. (a)(i) Adrenaline

OR

(a)(ii) Adrenal gland
(b) Heartbeat becomes faster.
- Breathing rate increases.

(c)

Hormonal Communication	**Electrical Impulses**
Chemical in nature	Electrical in nature
Carried through blood	Carried through neurons
Slower response	Very fast response
Effects are generally long-lasting	Effects are generally short-lasting
Can reach many target organs	Reach only nerve-connected cells

32 (a)

Plants	Animals
Hormones are synthesised in different plant tissues or regions.	Hormones are produced by specialised endocrine glands.
No specialised endocrine glands are present.	Specialised endocrine glands are present.
Hormones generally diffuse to their site of action.	Hormones are transported through blood.

(ii) (1) Intelligence → Cerebrum
(2) Riding a bicycle → Cerebellum
(3) Vomiting → Medulla Oblongata
(4) Controlling hunger → Hypothalamus (Forebrain)
(iii) Brain:
- Protected by the skull (cranium).
- Covered by three meninges.
- Surrounded by cerebrospinal fluid (CSF) which acts as a shock absorber.

Spinal Cord:
- Protected by the vertebral column (backbone).
- Also covered by meninges and surrounded by CSF.

OR

32 (b) (i) Tropic movements are directional growth movements of plants in response to external stimuli such as light, gravity, water or touch.
Example of plant hormones:
(1) Inhibits growth → Abscisic Acid (ABA)
(2) Promotes cell division → Cytokinin
When the tendril touches a support, auxin moves away from the side receiving the touch and accumulates on the opposite side.

Cells on the opposite side elongate faster than those on the touched side.
As a result, the tendril bends towards the support and coils around it.
Hormone responsible: Auxin

33 (a) (i) A reflex arc is the pathway followed by nerve impulses during a reflex action.
Reflex arcs have evolved in animals to provide a quick and automatic response to harmful stimuli, thereby protecting the body from injury without waiting for conscious instructions from the brain.
Sequence of events: Hot object (stimulus)
→ Receptors in the skin
→ Sensory neuron
→ Relay neuron in the spinal cord
→ Motor neuron
→ Muscles of the hand (effector)
→ Hand is withdrawn
(ii) The Peripheral Nervous System (PNS) helps in communication between the central nervous system and the other parts of the body.
Its two components are:

1. Cranial nerves arising from the brain
2. Spinal nerves arising from the spinal cord

OR

33 (b) (i)

- Stimulus: Touch
- Touch produces a signal that causes water to move out of the cells at the base of the leaves.
- Due to the loss of water, the cells lose their turgidity, causing the leaves to fold and droop.
- No growth is involved in this movement.
- It is a nastic movement and is temporary and reversible.

(ii) Geotropism is the directional growth movement of a plant part in response to gravity.

- Positive geotropism: Growth of a plant part towards gravity.
- Example: Roots grow downward towards gravity.

- Negative geotropism: Growth of a plant part away from gravity.
- Example: Shoots or stems grow upward, away from gravity.

34 (a) (i) Pituitary Gland
Hormone: Growth Hormone (GH)
Function: Promotes normal growth of bones and body tissues.
Deficiency Disorder: Dwarfism
(ii) Thyroid Gland
Hormone: Thyroxine
Function: Regulates the metabolism of carbohydrates, proteins and fats.
Deficiency Disorder: Goitre (due to iodine deficiency)
(b) Hormone secretion is regulated by a feedback mechanism.
Example: When blood glucose level increases, the pancreas secretes insulin.
Insulin lowers the blood sugar level.
Once the blood sugar returns to normal, insulin secretion decreases.
Thus, both the timing and amount of hormone secretion are regulated according to the body's needs.

35. Chemical coordination in plants occurs through plant hormones. These hormones are produced in different p

parts of the plant and diffuse to the region where they act. They control the growth, development and responses of plants to different stimuli.

Plant Hormone	Main Function
Auxin	Promotes cell elongation and helps the shoot bend towards light
Gibberellin	Promotes growth of the stem
Cytokinin	Promotes cell division
Abscisic acid (ABA)	Inhibits growth and causes wilting of leaves

CASE STUDY/SOURCE BASED QUESTION:

36. (a) Cerebrum
(b) Coordinates voluntary movements.
- Maintains balance and posture.
- Ensures smooth and precise movements.

(c) Conducts nerve impulses between the brain and body.
- Controls reflex actions.

OR

(c) The cerebrum helps in judging the speed, position and direction of the ball and controls the voluntary action. The cerebellum coordinates the muscular movements and ensures that the action is smooth and precise.

37. (a) Regulates blood glucose level.
- Helps body cells absorb glucose.
- Converts excess glucose into glycogen.

(b) Hormone secretion is regulated by a feedback mechanism. For example, when blood glucose increases, the pancreas secretes insulin. As blood glucose returns to normal, insulin secretion decreases.
(c)Required for the production of thyroxine.Prevents goitre.

OR

(c) Adrenaline (secreted by the adrenal gland).

38. (a) A synapse is the small gap between two neurons where nerve impulses are transmitted through chemical signals.
(b) Electrical impulse reaches the axon terminal.
- Neurotransmitters are released into the synapse.
- They cross the synapse and stimulate the next neuron.

(c) Chemicals are released from the axon ending of one neuron, cross the synapse and initiate an electrical impulse in the dendrite of the next neuron. Hence, transmission across the synapse occurs in one direction.

OR

(c) Receives information: Dendrites
- Transmits information: Axon

39. (a) Phototropism is the directional growth movement of a plant in response to light.
(b) Auxin accumulates on the shaded side of the shoot, causing greater cell elongation there. As a result, the shoot bends towards the light.
(c) Geotropism is the directional growth movement of a plant in response to gravity.
Example: Roots → Positive geotropism
Stem → Negative geotropism

OR

(c) Roots show positive hydrotropism by growing towards the region with more water.

40. (A) (2) — Cranium/skull
Function: It forms a bony box around the brain and protects it from mechanical injury.
(B) (i) Hypothalamus – Regulates body temperature (or controls thirst).
(ii) Cerebellum – It coordinates voluntary muscular movements during activities such as walking, riding a bicycle and picking up an object.
(C) (6) – Medulla oblongata
Functions:Controls involuntary actions like breathing and heartbeat.

- Controls reflexes such as swallowing, coughing and vomiting.

(D)

Basis	Part (1) – Cerebrum	Part (3) – Cerebellum
(i) Their size	It is the largest part of the brain.	It is the second largest part of the brain.
(ii) Functions	It controls thinking, memory, intelligence, reasoning, learning and voluntary actions. It also receives and interprets sensory information.	It coordinates muscular movements and maintains body balance and posture.

(E) Hindbrain: Cerebellum, Pons and Medulla oblongata.

41. (A) Figure A: The bending of the shoot towards light is called positive phototropism. Auxin moves to the shaded side of the shoot and causes the cells on that side to grow longer. Therefore, the shoot bends towards the light.

OR

Figure B: When the leaves of the touch-me-not plant are touched, an electrochemical signal causes changes in the amount of water present in the cells at the base of the leaves. These cells lose water and shrink, causing the leaves to fold and droop. This movement does not involve growth

(B) Compare the movement of growth of the pollen tube towards ovule with the movement shown in part A of the above figure.

Basis	Pollen tube towards ovule	Shoot bending towards light (Figure A)
1. Type of movement	Growth occurs due to chemotropism.	Growth occurs due to phototropism.
2. Stimulus	Stimulus is chemical substances released by the ovule.	Stimulus is light.
3. Nature of movement	It is a directional growth movement.	It is also a directional growth movement.

(C) Compare the movement shown in figure B with the movement of body parts in the animals.

Basis	Figure B (Touch-me-not plant)	Movement in animals
1. Cause of movement	Movement occurs due to changes in turgor pressure in the cells.	Movement occurs due to contraction and relaxation of muscles.
2. Involvement of muscles	Does not involve muscles.	Involves muscles.
3. Control	It is a nastic movement in response to touch.	It is controlled by the nervous system and may be voluntary or involuntary.